Agentic Document Processor

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Model Selection

Classifier Model

llama-3.1-8b-instant

Extractor Model

openai/gpt-oss-120b

Redactor Model

llama-3.3-70b-versatile

Process Document

Extracted JSON


```
{
  "invoice_details" : {
    "invoice_number" : "NP#00183"
    "invoice_date" : "24.06.2020"
    "due_date" : "24.09.2020"
    "purchase_order_number" : NULL
```

```
    "currency" : NULL
  }
  "vendor" : {
    "name" : "notapino"
    "address" : "Ria de Goa 1 Amadora, Lisbon, 2720-059"
    "phone" : "+351 21 474 1957"
    "email" : "contact@notapino.pt"
    "tax_id" : "PT285637271"
  }
  "bill_to" : {
    "name" : "Actoonda"
    "address" : "Rua Godinho 634 Matosinhos, Porto, 4450-147"
    "phone" : "+351 22 938 0548"
    "email" : "office@actoonda.pt"
  }
  "line_items" : [
    0 : {
      "description" : "Skid-Steer 822-79-9581"
      "quantity" : "5."
      "unit_price" : "200"
      "total" : "1000"
    }
    1 : {
      "description" : "Backhoe I 598-31-5899"
      "quantity" : "5."
      "unit_price" : "500"
      "total" : "1000"
    }
    2 : {
      "description" : "Dump Truck I 745-73-2437"
      "quantity" : "2."
      "unit_price" : "1500"
      "total" : "3000"
    }
    3 : {
```

```
"description" : "Trencher I 813-54-6535"
"quantity" : "4."
"unit_price" : "300"
"total" : "1200"
}
4 : {
"description" : "Grader 113-98-4866"
"quantity" : "10."
"unit_price" : "200"
"total" : "2000"
}
5 : {
"description" : "Dump Truck II"
"quantity" : NULL
"unit_price" : NULL
"total" : "10000"
}
6 : {
"description" : "Bulldozer I"
"quantity" : "5."
"unit_price" : "2000"
"total" : "750"
}
7 : {
"description" : "Scraper I"
"quantity" : "5."
"unit_price" : "200"
"total" : "200"
}
8 : {
"description" : "Compactor I"
"quantity" : "5."
"unit_price" : "250"
"total" : "350"
}
```

```
9 : {
  "description" : "Scraper II"
  "quantity" : "5."
  "unit_price" : "300"
  "total" : "450"
}
10 : {
  "description" : "Crawler I"
  "quantity" : "10."
  "unit_price" : "600"
  "total" : "500"
}
11 : {
  "description" : "Compactor II"
  "quantity" : "4."
  "unit_price" : "600"
  "total" : "800"
}
12 : {
  "description" : "Crawler II"
  "quantity" : "2."
  "unit_price" : "1000"
  "total" : "650"
}
13 : {
  "description" : "Backhoe II"
  "quantity" : "4."
  "unit_price" : "1250"
  "total" : "900"
}
14 : {
  "description" : "Dragline"
  "quantity" : "4."
  "unit_price" : "3750"
  "total" : "1000"
```

```
        }
        15 : {
          "description" : "Trencher II"
          "quantity" : "6."
          "unit_price" : "1000"
          "total" : "2500"
        }
        16 : {
          "description" : "Backhoe III"
          "quantity" : "5."
          "unit_price" : "1400"
          "total" : "600"
        }
        17 : {
          "description" : "Bulldozer II"
          "quantity" : "5."
          "unit_price" : "1800"
          "total" : "2400"
        }
        18 : {
          "description" : "Crawler III"
          "quantity" : "10."
          "unit_price" : "3000"
          "total" : "3000"
        }
    ]
}
```


Redacted JSON


```
{
  "invoice_details" : {
    "invoice_number" : "NP#00183"
    "invoice_date" : "24.06.2020"
    "due_date" : "24.09.2020"
    "purchase_order_number" : NULL
```

```
    "currency" : NULL
  }
  "vendor" : {
    "name" : "[NAME]"
    "address" : "[ADDRESS]"
    "phone" : "[PHONE]"
    "email" : "[EMAIL]"
    "tax_id" : "[ID]"
  }
  "bill_to" : {
    "name" : "[NAME]"
    "address" : "[ADDRESS]"
    "phone" : "[PHONE]"
    "email" : "[EMAIL]"
  }
  "line_items" : [
    0 : {
      "description" : "Skid-Steer 822-79-9581"
      "quantity" : "5."
      "unit_price" : "200"
      "total" : "1000"
    }
    1 : {
      "description" : "Backhoe I 598-31-5899"
      "quantity" : "5."
      "unit_price" : "500"
      "total" : "1000"
    }
    2 : {
      "description" : "Dump Truck I 745-73-2437"
      "quantity" : "2."
      "unit_price" : "1500"
      "total" : "3000"
    }
    3 : {
```

```
"description" : "Trencher I 813-54-6535"
"quantity" : "4."
"unit_price" : "300"
"total" : "1200"
}
4 : {
"description" : "Grader 113-98-4866"
"quantity" : "10."
"unit_price" : "200"
"total" : "2000"
}
5 : {
"description" : "Dump Truck II"
"quantity" : NULL
"unit_price" : NULL
"total" : "10000"
}
6 : {
"description" : "Bulldozer I"
"quantity" : "5."
"unit_price" : "2000"
"total" : "750"
}
7 : {
"description" : "Scraper I"
"quantity" : "5."
"unit_price" : "200"
"total" : "200"
}
8 : {
"description" : "Compactor I"
"quantity" : "5."
"unit_price" : "250"
"total" : "350"
}
```

```
9 : {
  "description" : "Scraper II"
  "quantity" : "5."
  "unit_price" : "300"
  "total" : "450"
}
10 : {
  "description" : "Crawler I"
  "quantity" : "10."
  "unit_price" : "600"
  "total" : "500"
}
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  "description" : "Compactor II"
  "quantity" : "4."
  "unit_price" : "600"
  "total" : "800"
}
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  "description" : "Crawler II"
  "quantity" : "2."
  "unit_price" : "1000"
  "total" : "650"
}
13 : {
  "description" : "Backhoe II"
  "quantity" : "4."
  "unit_price" : "1250"
  "total" : "900"
}
14 : {
  "description" : "Dragline"
  "quantity" : "4."
  "unit_price" : "3750"
  "total" : "1000"
```

```
        }
        15 : {
          "description" : "Trencher II"
          "quantity" : "6."
          "unit_price" : "1000"
          "total" : "2500"
        }
        16 : {
          "description" : "Backhoe III"
          "quantity" : "5."
          "unit_price" : "1400"
          "total" : "600"
        }
        17 : {
          "description" : "Bulldozer II"
          "quantity" : "5."
          "unit_price" : "1800"
          "total" : "2400"
        }
        18 : {
          "description" : "Crawler III"
          "quantity" : "10."
          "unit_price" : "3000"
          "total" : "3000"
        }
    ]
}
```


Redacted metrics


```
{
  "tp" : 9
  "fp" : 0
  "fn" : 0
  "precision" : 1
  "recall" : 1
```

```
  "f1" : 1
  "total_expected" : 9
  "total_predicted" : 9
}
```


Token_usage


```
[
  0 : {
    "classifier_agent" : {
      "input_tokens" : 478
      "output_tokens" : 95
      "total_tokens" : 573
    }
    "model_name" : "llama-3.1-8b-instant"
  }
  1 : {
    "extractor_agent" : {
      "input_tokens" : 2254
      "output_tokens" : 4096
      "total_tokens" : 6350
      "output_token_details" : {
        "reasoning" : 3133
      }
    }
    "model_name" : "openai/gpt-oss-120b"
  }
  2 : {
    "redactor_agent" : {
      "input_tokens" : 1278
      "output_tokens" : 873
      "total_tokens" : 2151
    }
    "model_name" : "llama-3.3-70b-versatile"
  }
]
```

Self Repair


```
{
  "attempts" : 0
  "last_trace" : NULL
}
```


Agent Traces


```
[
  0 : {
    "timestamp" : "2026-02-20T03:16:45.960612"
    "agent" : "ClassifierAgent"
    "input" : {
      "document_signals" : {
        "financial" : [
          0 : "monetary values"
          1 : "Due-date: 24.09.2020"
          2 : "Invoice number: NP#00183"
          3 : "Date: 24.06.2020"
        ]
        "logistics" : [
          0 : "shipping info"
          1 : "billing info"
        ]
        "correspondence" : [
          0 : "recipient"
          1 : "sender"
        ]
        "compliance" : [
          0 : "invoice"
          1 : "tax ID"
        ]
        "technical" : [
          0 : "product descriptions"
        ]
```

```
        "other" : [
          0 : "product list"
        ]
      }
      "entities" : {
        "organizations" : [
          0 : "notapino"
          1 : "Actoonda"
        ]
        "locations" : [
          0 : "Rua Godinho 634, Matosinhos, Porto, 4450-147"
          1 : "Ria de Goa 1, Amadora, Lisbon, 2720-059"
        ]
      }
      "dates" : [
        0 : "24.06.2020"
        1 : "24.09.2020"
      ]
      "identifiers" : [
        0 : "PT285637271"
        1 : "NP#00183"
      ]
      "monetary_values" : [
        0 : "€ 300"
        1 : "€ 10000"
        2 : "€ 3000"
        3 : "€ 1500"
        4 : "€ 2000"
        5 : "€ 1200"
        6 : "€ 1000"
        7 : "€ 500"
        8 : "€ 200"
      ]
      "pii_detected" : false
      "contains_signature" : false
      "contains_table" : true
```

```
"avg_chunk_confidence" : 0.9
"chunk_count" : 1
}
"output" : {
"predicted_document_type" : "invoice"
"confidence" : 0.9
}
"metadata" : {
"reasoning" :
"The presence of 'invoice number' and 'tax ID' in the 'compliance'
section, along with 'monetary values' and 'due-date' in the 'financial'
section, strongly suggests that the document is an invoice. This
conclusion is further supported by the absence of keywords typically
found in contracts, resumes, or other documents."
}
}
1 : {
"timestamp" : "2026-02-20T03:16:55.567576"
"agent" : "ExtractorAgent"
"input" : {
"doc_type" : "invoice"
"context_preview" :
"INVOICE
notapino
Ria de Goa 1
Amadora, Lisbon, 2720-059
+351 21 474 1957
contact@notapino.pt
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod
tempor incididunt ut labore et dolore magna aliqua. Leo integer
malesuada nunc vel risus. Elit ut aliquam purus sit amet luctus. Nibh
cras pulvinar mattis nunc sed blandit libero. Congue nisi vitae
suscipit tellus mauris a diam. Iaculis eu non diam phasellus vestibulum
lorem sed risus. Dictumst quisque sagittis purus sit amet. Neque "
}
"output" : {
"confidence" : NULL
"top_level_fields" : [
0 : "invoice_details"
```

```
            1 : "vendor"
            2 : "bill_to"
            3 : "line_items"
          ]
        }
        "metadata" : {
          "model" : "openai/gpt-oss-120b"
          "extracted_field_count" : 4
        }
      }
      2 : {
        "timestamp" : "2026-02-20T03:16:57.663379"
        "agent" : "RedactAgent"
        "input" : {
          "extracted_preview" :
          "{'invoice_details': {'invoice_number': 'NP#00183', 'invoice_date':
          '24.06.2020', 'due_date': '24.09.2020', 'purchase_order_number': None,
          'currency': None}, 'vendor': {'name': 'notapino', 'address': 'Ria de
          Goa 1 Amadora, Lisbon, 2720-059', 'phone': '+351 21 474 1957', 'email':
          'contact@notapino.pt', 'tax_id': 'PT285637271'}, 'bill_to': {'name':
          'Actoonda', 'address': 'Rua Godinho 634 Matosinhos, Porto, 4450-147',
          'phone': '+351 22 938 0548', 'email': 'office@actoonda.pt'},
          'line_items': [{'desc"
          "doc_type" : "invoice"
        }
        "output" : {
          "redacted_preview" :
          "{'invoice_details': {'invoice_number': 'NP#00183', 'invoice_date':
          '24.06.2020', 'due_date': '24.09.2020', 'purchase_order_number': None,
          'currency': None}, 'vendor': {'name': '[NAME]', 'address': '[ADDRESS]',
          'phone': '[PHONE]', 'email': '[EMAIL]', 'tax_id': '[ID]'}, 'bill_to':
          {'name': '[NAME]', 'address': '[ADDRESS]', 'phone': '[PHONE]', 'email':
          '[EMAIL]'}, 'line_items': [{'description': 'Skid-Steer 822-79-9581',
          'quantity': '5.', 'unit_price': '200', 'total': '1000'},
          {'description': 'Backh"
          "pii_metrics" : {
            "tp" : 9
            "fp" : 0
```

```
            "fn" : 0
            "precision" : 1
            "recall" : 1
            "f1" : 1
            "total_expected" : 9
            "total_predicted" : 9
          }
          "mask_counts" : {
            "[NAME]" : 2
            "[EMAIL]" : 2
            "[PHONE]" : 2
            "[ADDRESS]" : 2
            "[ID]" : 1
            "[BANK]" : 0
            "[DOB]" : 0
            "[CARD]" : 0
            "[SIGNATURE]" : 0
          }
        }
        "metadata" : {
          "attempt" : 1
        }
      }
]
```


Latency

Processing Time (seconds)

15.131

Latency Breakdown

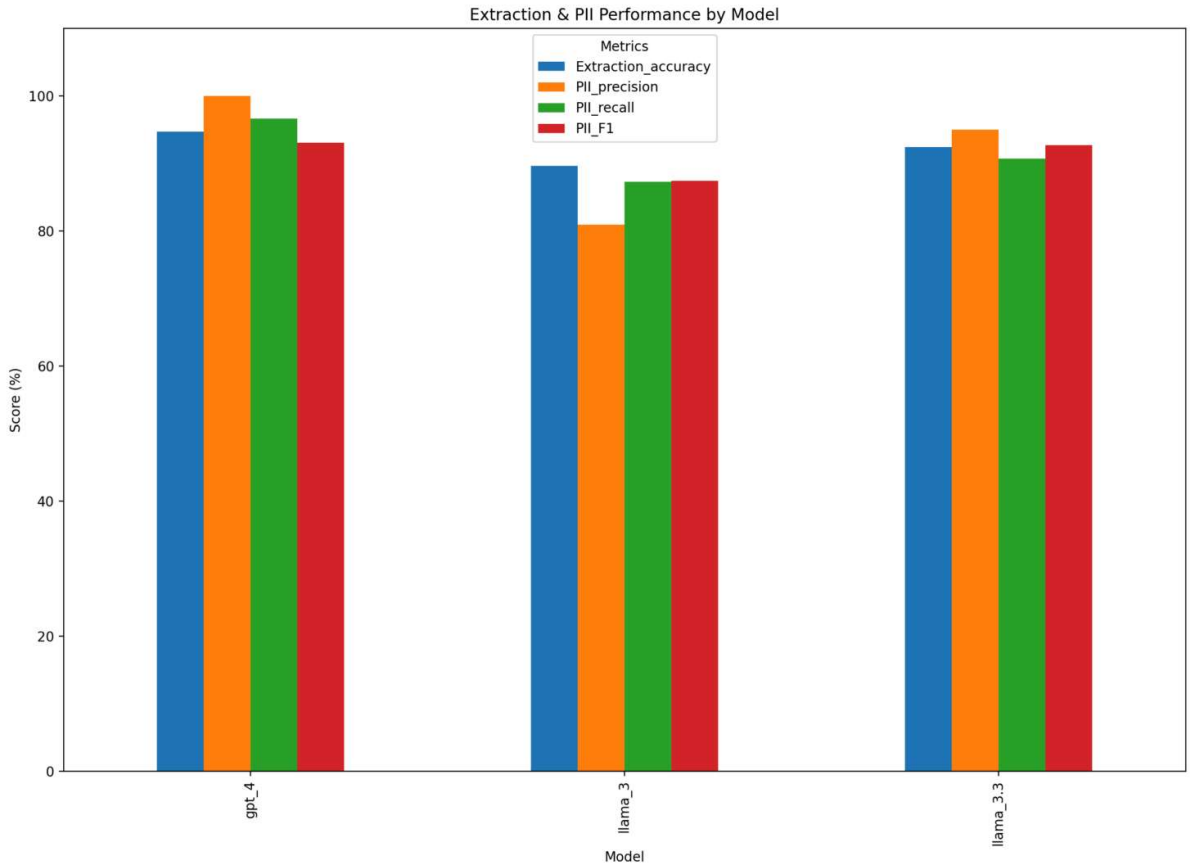

```
{
  "ingestion" : 0.385
  "chunking" : 0.001
```

```
  "summarization" : 1.926
  "aggregation" : 0
  "graph_execution" : 12.818
}
```


Model valuation

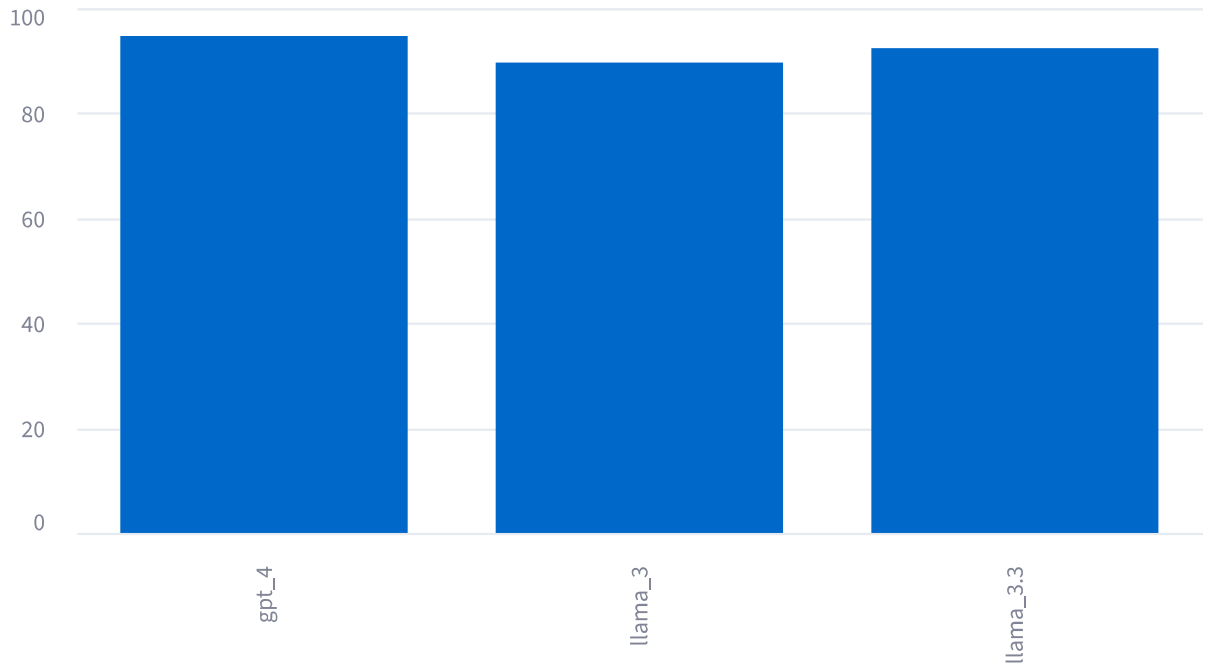
model_name	Extraction_accuracy	PII_precision	PII_recall	PII_F1
gpt_4	94.71	100	96.62	93.06
llama_3	89.62	80.88	87.22	87.38
llama_3.3	92.37	95	90.66	92.66

Model Performance Comparison

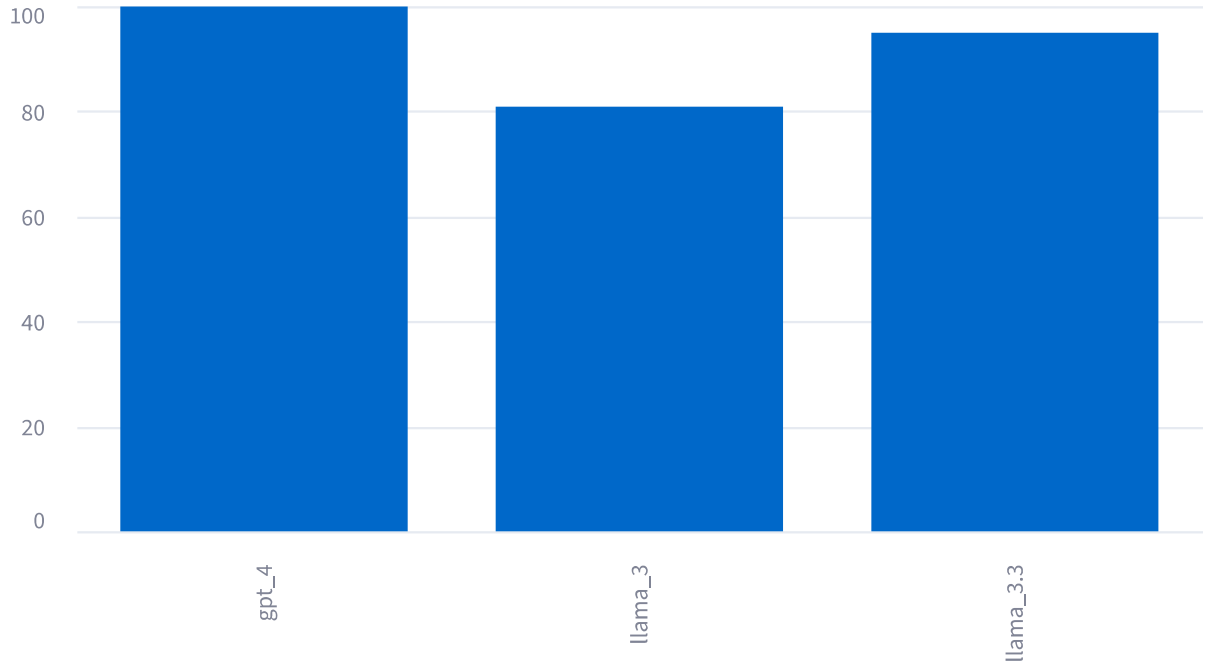


Model Performance Breakdown

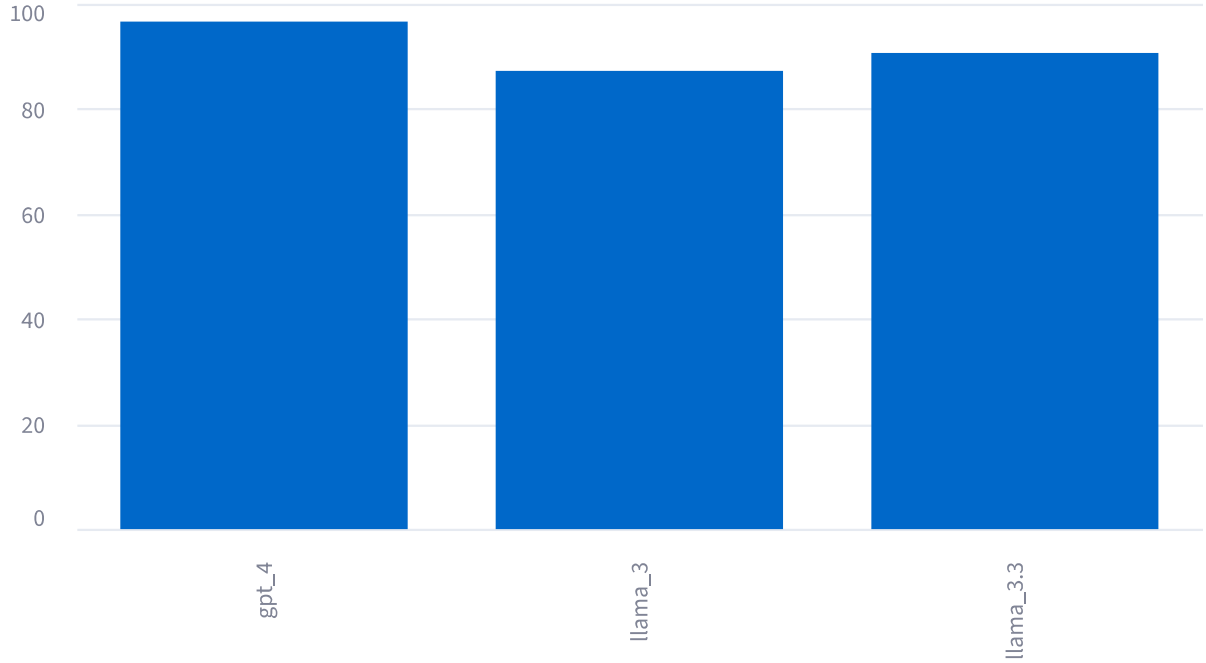
Extraction_accuracy



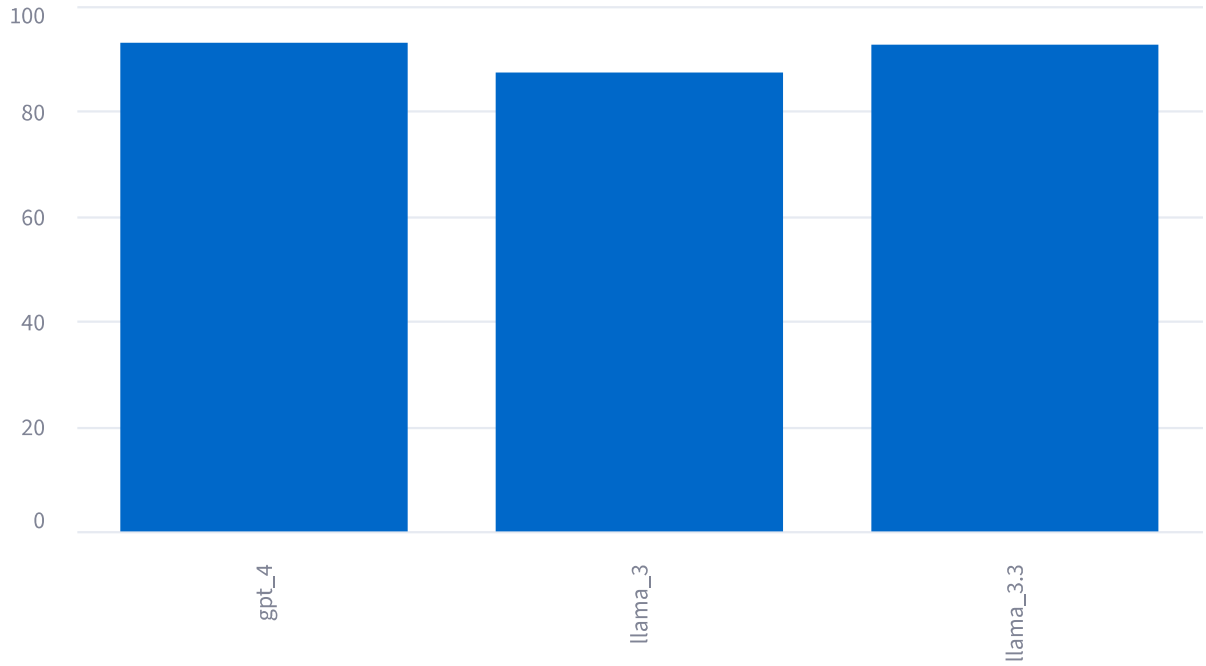
PII_precision



PII_recall



PII_F1



🏆 Quick Insights From Model Evaluation

🥇 **Best Extraction Accuracy → gpt_4** (94.71%) 🥇 **Best PII Precision → gpt_4** (100.00%) 🥇 **Best PII Recall → gpt_4** (96.62%) 🥇 **Best F1 Overall → gpt_4** (93.06%)